

DSA PROBLEM NOTES

Leaders of an Array

1. PROBLEM IN MY WORDS

An element is a leader if it is greater than every element to its right. The rightmost element is always a leader because there is nothing to its right.

2. EXAMPLE

Input	[16, 17, 4, 3, 5, 2]
Leaders	[17, 5, 2]
Why?	17 is greater than everything to its right; 5 is greater than 2; 2 is the last element.

3. BRUTE FORCE

For every element, check all elements to its right. If no right-side element is greater, it is a leader.

Time: $O(n^2)$

Space: $O(1)$ excluding the output list.

4. PATTERN □

Reverse Traversal + Maximum So Far

Why?

A leader depends only on the elements to its right. Traverse from right to left while keeping the maximum value seen so far.

5. CORE IDEA □

1. The last element is automatically a leader.
2. Keep maxRight, the largest value seen on the right.
3. Move from right to left.
4. If $\text{nums}[i] > \text{maxRight}$, $\text{nums}[i]$ is a leader.
5. Update maxRight whenever a larger value is found.

This reduces the solution from $O(n^2)$ to $O(n)$.

6. DRY RUN

```
nums = [16, 17, 4, 3, 5, 2]
```

2 → leader → maxRight = 2

5 → 5 > 2 → leader → maxRight = 5

3 → 3 < 5 → not leader

4 → 4 < 5 → not leader

17 → 17 > 5 → leader → maxRight = 17

16 → 16 < 17 → not leader

Found right-to-left: [2, 5, 17]

Final answer: [17, 5, 2]

7. CODE

```
import java.util.*;

class Solution {
    public static ArrayList<Integer> leaders(int[] nums) {
        ArrayList<Integer> ans = new ArrayList<>();

        int maxRight = nums[nums.length - 1];
        ans.add(maxRight);

        for (int i = nums.length - 2; i >= 0; i--) {
            if (nums[i] > maxRight) {
                ans.add(nums[i]);
                maxRight = nums[i];
            }
        }

        Collections.reverse(ans);
        return ans;
    }
}
```

8. COMPLEXITY

Time Complexity: $O(n)$

Space Complexity: $O(n)$ for the output list.

Extra algorithmic space apart from the output is $O(1)$.

9. MISTAKES ⚠

- The rightmost element is always a leader.
- Use $>$, not $>=$, for a strictly greater definition.
- Leaders are discovered in reverse order.
- Reverse the result if the expected answer is left-to-right.

10. RECOGNITION CLUE 🧠

Condition depends on all elements to the right → think Reverse Traversal.

If you need information about the right side, try scanning right-to-left and maintaining a maximum/minimum so far.

11. RELATED PROBLEMS

- Next Greater Element
- Stock Span / monotonic stack problems
- Suffix Maximum / Suffix Minimum problems
- Trapping Rain Water